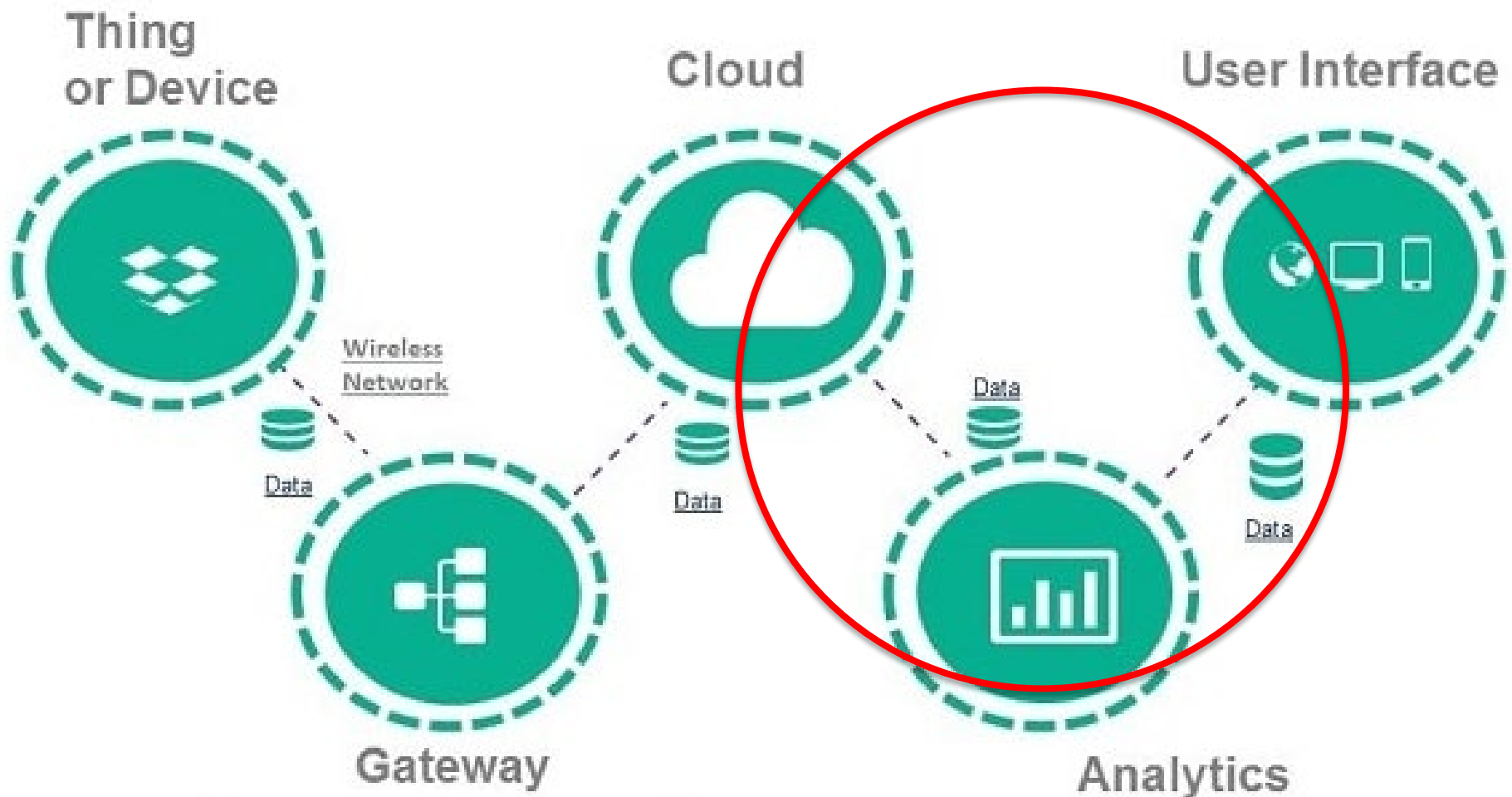


Industrial Internet of Things for Advanced Manufacturing

IoT Cloud Stack
Data Storage and Visualization



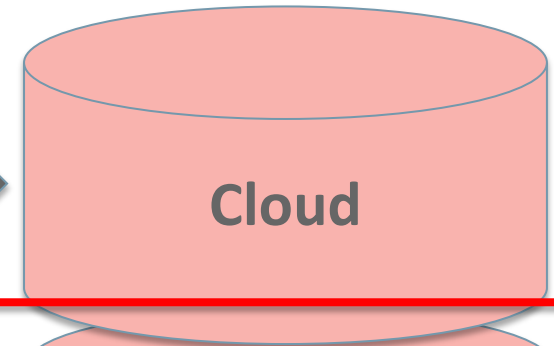
Recall: Building blocks of IoT



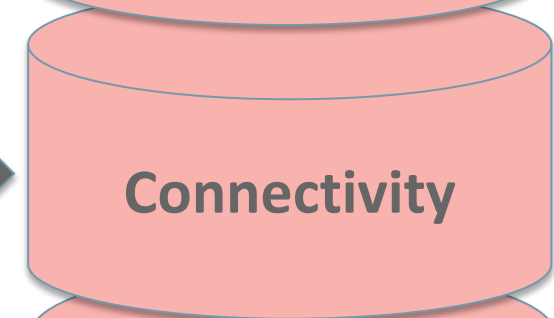
<https://www.rfpage.com/what-are-the-major-components-of-internet-of-things/>

Recall: IoT Stack vs. Subject Aims

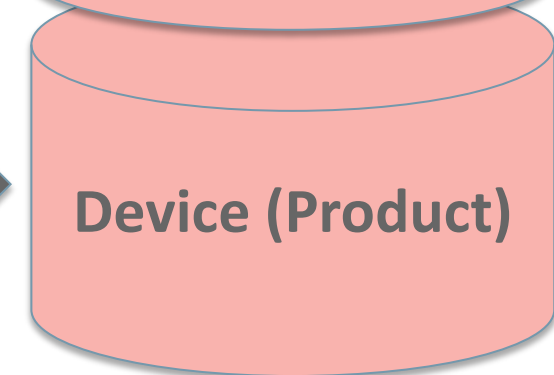
configure visualisation platform for monitoring of data from the sensor network.



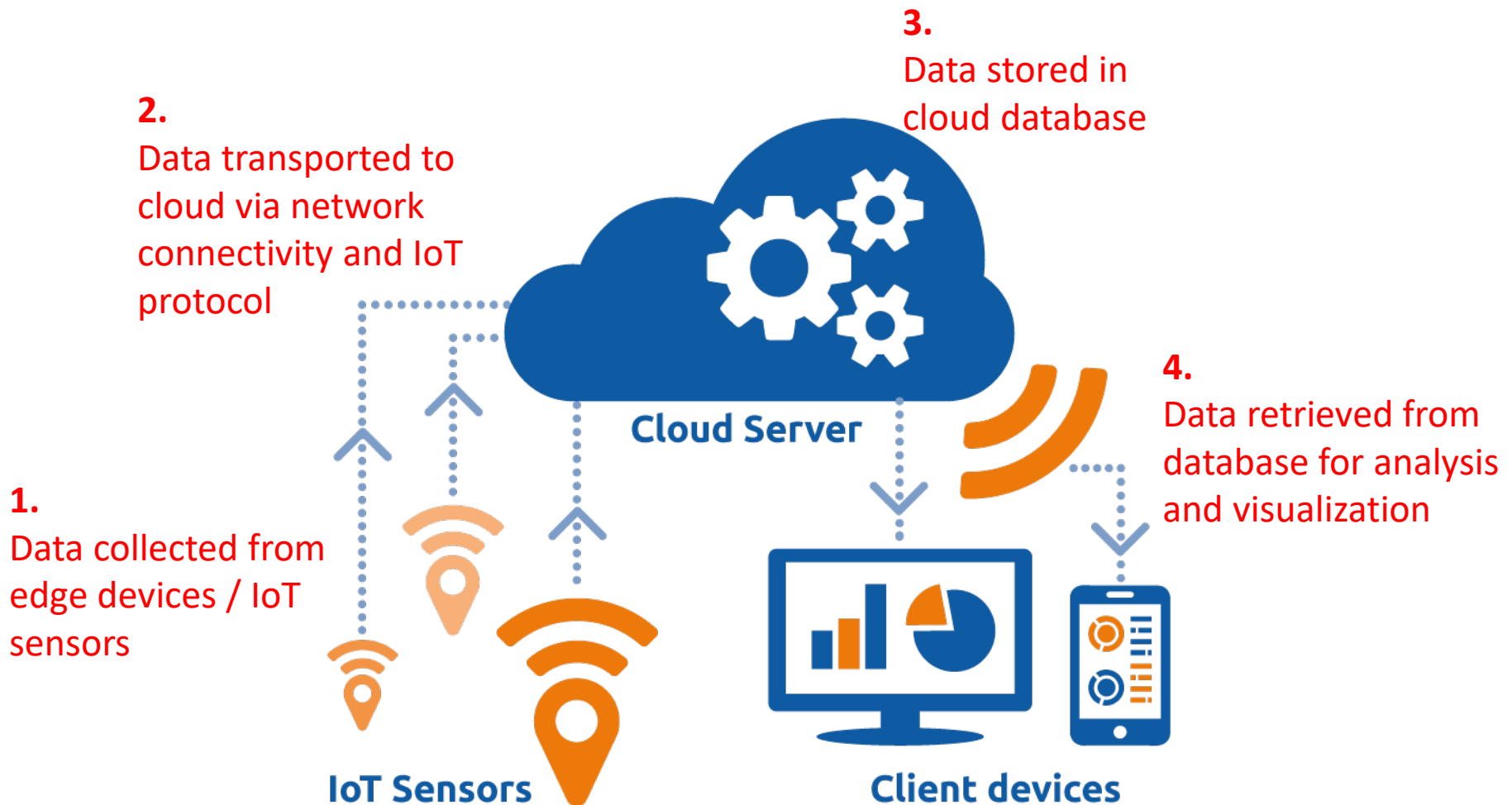
configure embedded devices for network connectivity and apply data transport protocols to enable data communication with cloud services.



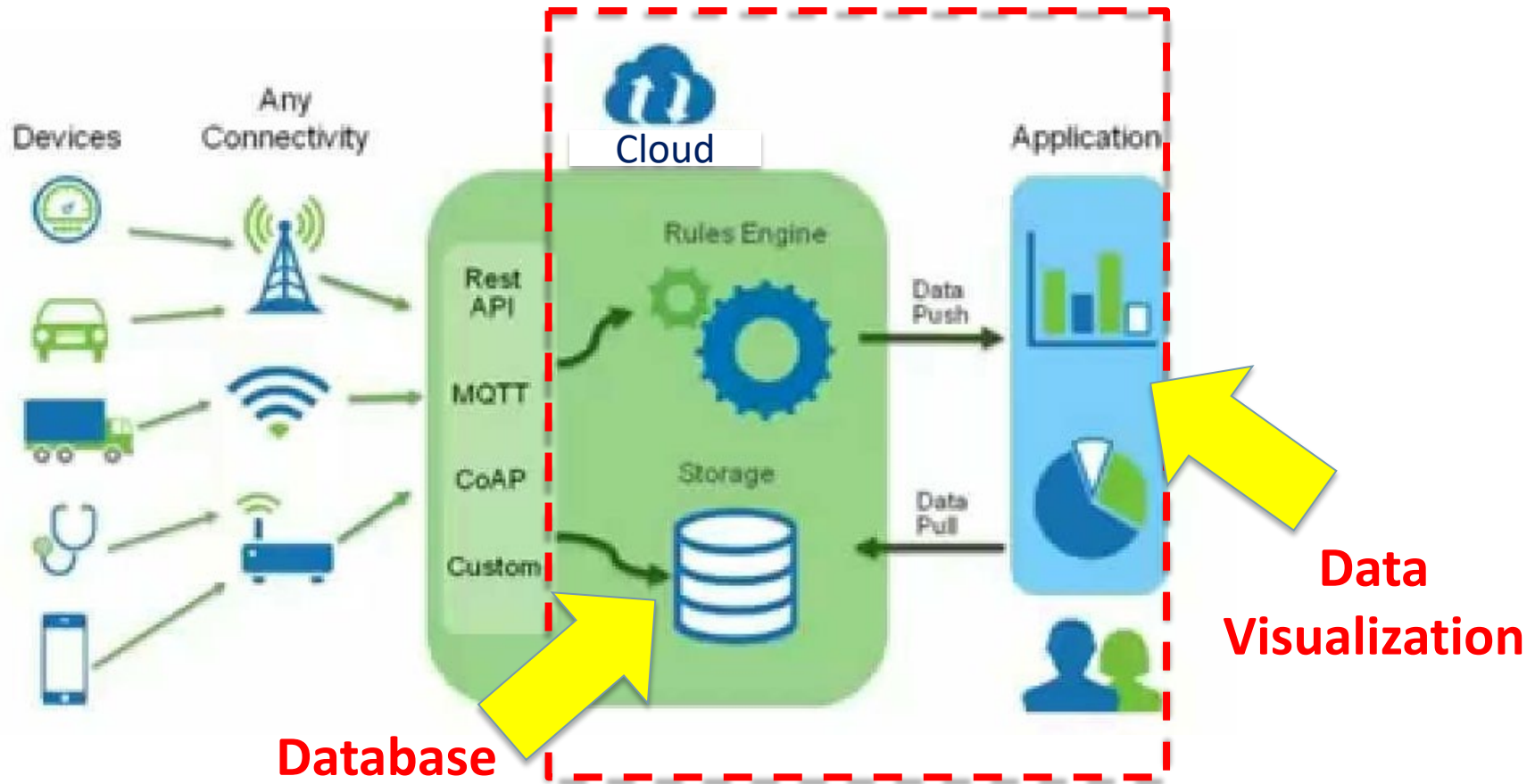
develop applications to enable sensing and control in embedded devices.



Recall: Concept of IoT



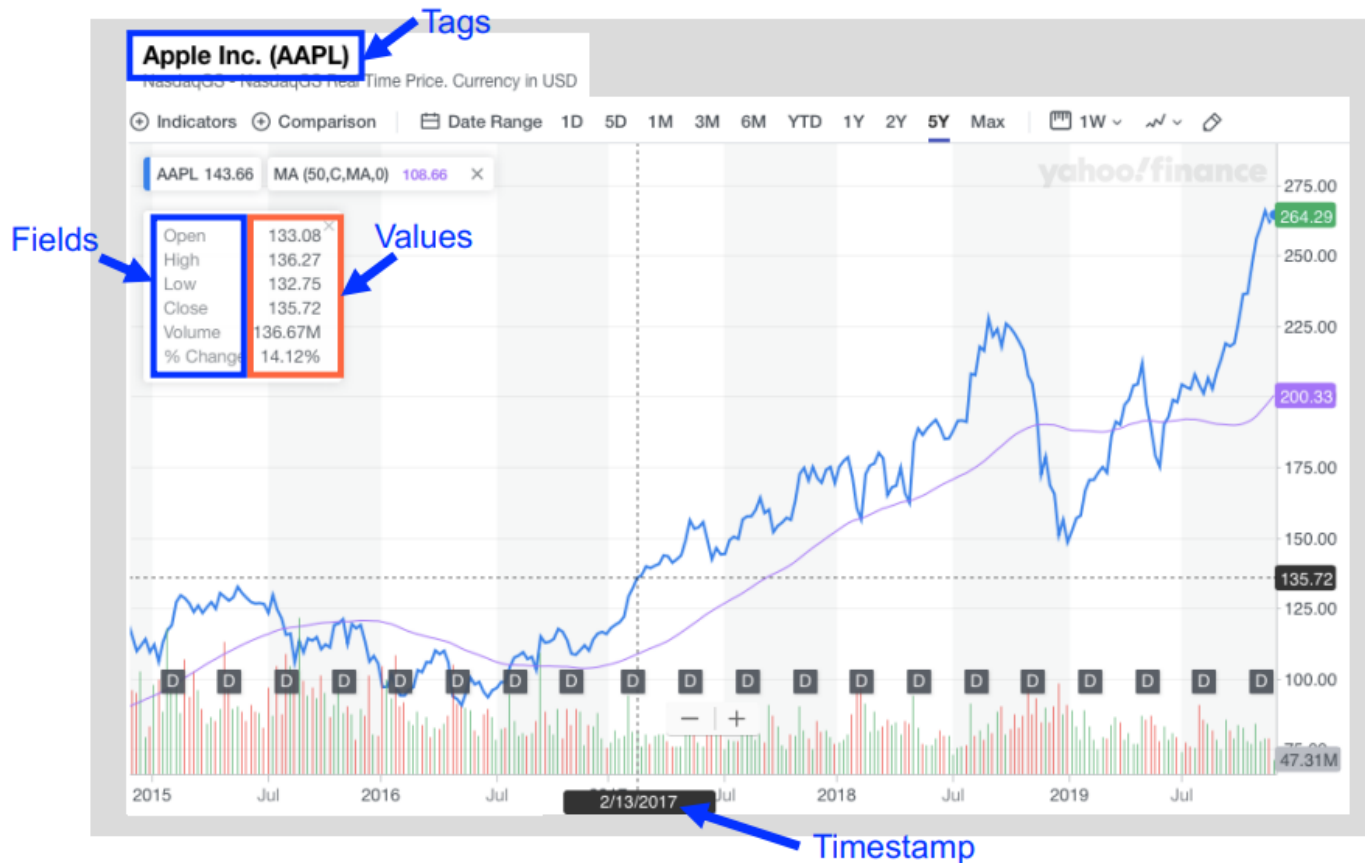
Recall: IoT Architecture



Time Series Data

- Time series data is any data that has a timestamp, i.e. data collected at regular intervals over a period of time.
- Some examples of time series data:
 - ✓ IoT device (sensor) data
 - ✓ Ocean tides
 - ✓ Stocks
 - ✓ Commodity prices
- Rapid growth of IoT and sensor-based data have led to a need for scale, performance and high availability which traditional relational structured database cannot support.

Time Series Data



Measurement	Stock_Price
Tag	Name=Apple Inc. Symbol=AAPL
Fields	Open=133.08 High=136.27 Low=132.75 Close=135.72 Volume=136.67M Change=14.12%
Timestamp	2/13/2017

Time Series Database

- It offers the scalability and performance that traditional relational databases are unable to offer, particularly to meet the needs of IoT data today.
- Some examples of time series database:
 - ✓ Atlas
 - ✓ Druid
 - ✓ InfluxDB (lab 6)
 - ✓ NoSQL

Data Visualization

- The goals of data visualization are to:
 - ✓ communicate information clearly and effectively so as to help user understand the significance of data through graphical means.
 - ✓ study patterns, trends and correlations that might go undetected in text-based data
- The eventual aim is to make intelligent and actionable recommendations for improvement in systems and processes.

Data Visualization Tools

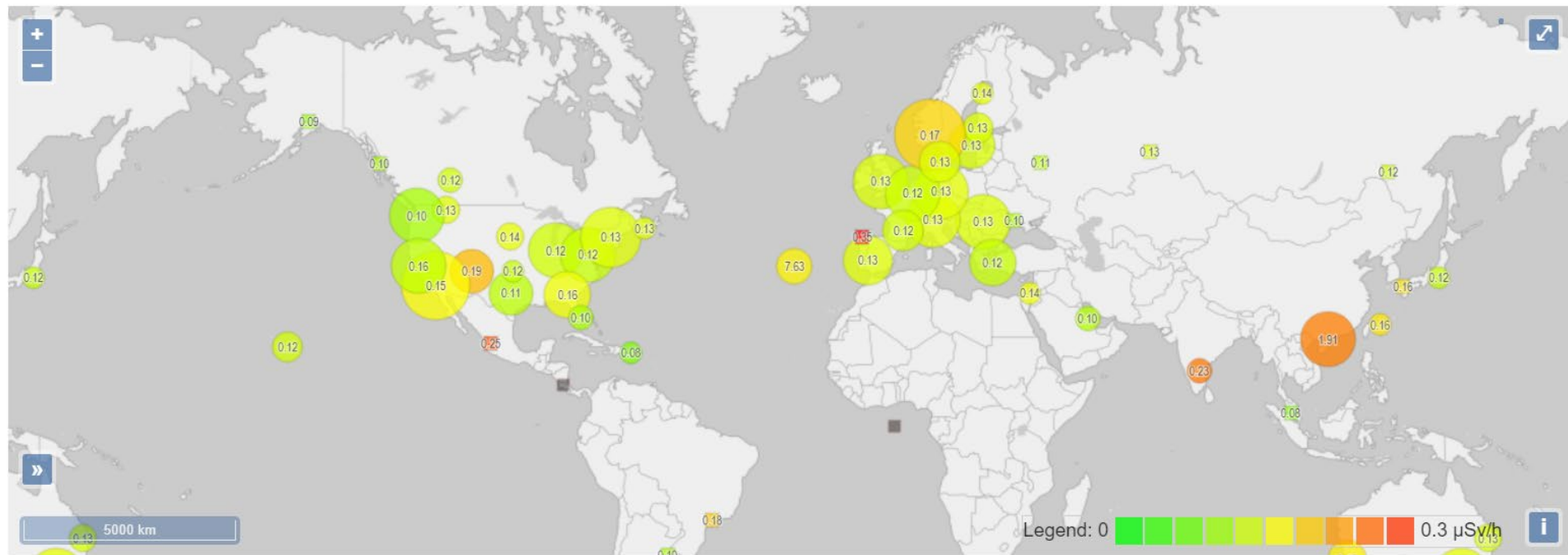
- Some examples of data visualization tools:
 - ✓ Tableau
 - ✓ Fusion Charts
 - ✓ Google Charts
 - ✓ Excel
 - ✓ R
 - ✓ Chronograf
 - ✓ Grafana (lab 6)

Briefing on Lab 6:

Data Storage and Visualization

IoT Case Study: Global Environmental Monitoring

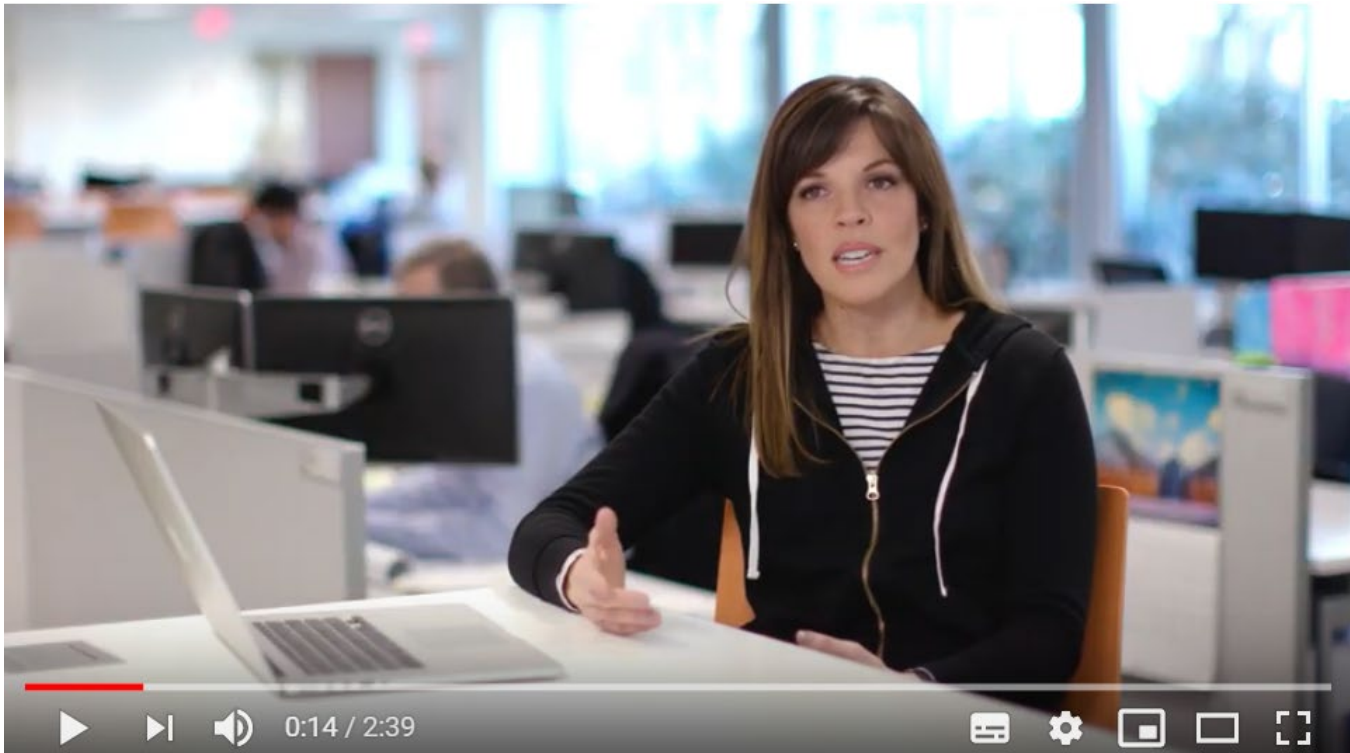
Spend 3 minutes to read the case and share your thoughts.



<https://cleantechnica.com/2015/12/11/uradmonitor-global-pollution-tracking-network/>
<https://www.uradmonitor.com/>

Why develop on an IoT platform ?

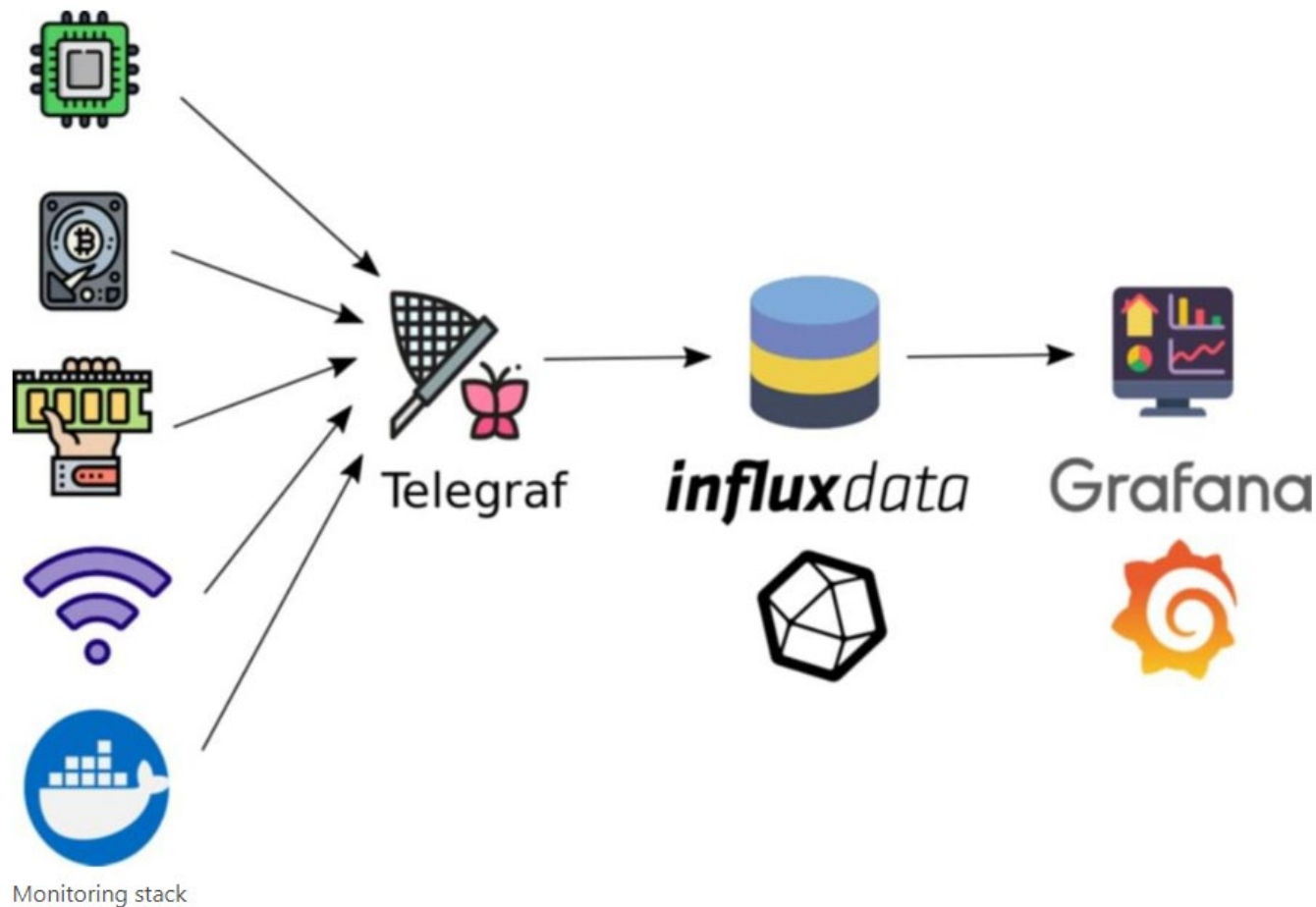
A platform is a software and hardware environment under which various smaller application programs can be developed and run.



What we need ?

- Collect data
- Store and process data
- Visualize data
- Analytics
- Monitoring and alert

Modern Monitoring and Process Architecture



Why InfluxDB?

□ include secondary database models

43 systems in ranking, September 2023

Rank			DBMS	Database Model	Score		
Sep 2023	Aug 2023	Sep 2022			Sep 2023	Aug 2023	Sep 2022
1.	1.	1.	InfluxDB 	Time Series, Multi-model 	31.27	+1.40	+2.12
2.	2.	2.	Kdb 	Multi-model 	8.94	+0.52	+0.81
3.	3.	 4.	Prometheus	Time Series	7.62	-0.22	+1.09
4.	4.	 3.	Graphite	Time Series	5.45	-0.15	-1.11
5.	5.	5.	TimescaleDB	Time Series, Multi-model 	5.39	+0.28	+0.53
6.	6.	 9.	DolphinDB	Time Series, Multi-model 	4.04	+0.55	+2.42
7.	 8.	 6.	Apache Druid	Multi-model 	3.20	-0.10	+0.57
8.	 7.	 7.	RRDtool	Time Series	3.14	-0.16	+0.62
9.	9.	 13.	TDengine 	Time Series, Multi-model 	2.62	-0.06	+1.31
10.	10.	 11.	QuestDB 	Time Series, Multi-model 	2.38	-0.14	+0.87

Source: <https://db-engines.com/en/ranking/time+series+dbms>

What is Grafana?

- Open source software for time series analytics
- Advanced, visualize and rich metrics dashboard
- Reporting and monitoring server
- Alert Notification feature
- Widely adopted

